

Question :

Write a program to implement First Come First Serve CPU scheduling in C.

Code :

```
#include <stdio.h>

int main()
{
    int n = 5;

    int pid[] = {0, 1, 2, 3, 4};
    int bt[] = {2, 6, 4, 9, 12};
    int at[] = {0, 1, 2, 3, 4};

    int ct[5], tat[5], wt[5];

    int total_tat = 0;
    int total_wt = 0;

    // FCFS Scheduling

    for (int i = 0; i < n; i++)
    {
        if (i == 0)
        {
```

```
        ct[i] = at[i] + bt[i];
    }
    else
    {
        if (ct[i - 1] < at[i])
            ct[i] = at[i] + bt[i];
        else
            ct[i] = ct[i - 1] + bt[i];
    }

    // Turn Around Time = Completion Time - Arrival Time
    tat[i] = ct[i] - at[i];

    // Waiting Time = Turn Around Time - Burst Time
    wt[i] = tat[i] - bt[i];

    total_tat += tat[i];
    total_wt += wt[i];
}
```

```

// Display Table

printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

for (int i = 0; i < n; i++)
{
    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
           pid[i], at[i], bt[i],
           ct[i], tat[i], wt[i]);
}

printf("\nAverage Turn Around Time = %.2f",
       (float)total_tat / n);

printf("\nAverage Waiting Time = %.2f\n",
       (float)total_wt / n);

return 0;
}

```

Output :

```

dev@SOHAM-PC:~/Education/Operating-Systems-Programs/Classroom$ gcc FCFS.c && ./a.out
PID    AT     BT     CT     TAT    WT
0      0      2      2      2      0
1      1      6      8      7      1
2      2      4      12     10     6
3      3      9      21     18     9
4      4      12     33     29     17

Average Turn Around Time = 13.20
Average Waiting Time = 6.60

```